



Key Blocks

Frequently Asked Questions

31 August 2022

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Document Purpose

The purpose of this document is to answer commonly asked questions about Mastercard's changes to support the Payment Card Industry (PCI) PIN Security Requirements and Testing Procedures involving key exchange information storage using key blocks.

Reference Documents

Refer to the following reference manuals available on [Mastercard Connect™](#) in the Technical Resource Center (TRC) for information about the current state of Mastercard processing:

- *Customer Interface Specification*
- *Single Message System Specifications*

Announcements

Refer to the following published announcements for more information on Key Blocks:

- [AN 2944—Update to the Announced Cryptographic Key Block Changes Supporting Phase 2 of PCI Mandates](#)
- [AN 2117—Introducing Cryptographic Key Block Changes to Support Phase 2 of PCI Mandates](#)
- *X9 TR-31 Interoperable Secure Key Exchange Key Block Specification for Symmetric Algorithms, an industry standard, available from the [American National Standards Institute \(ANSI\) Webstore](#)*
- *Cryptographic Key Blocks, June 2017, an industry standard, available from the [PCI Security Standards Council \(PCI SSC\)](#)*

Key Blocks

The following list of questions is designed to assist customers and their agents with Mastercard's compliance requirements for key block implementations.

Q. What are Key Blocks?

Key Blocks is a standard way of protecting the integrity of cryptographic keys and of associating them with their intended use. They may be used to protect both Triple Data Encryption Algorithm (TDEA), sometimes referred to as 3DES or TripleDES (TDES), and Advanced Encryption Standard (AES) keys.

Q. How do Key Blocks improve the security of encrypted keys?

Key Blocks improve the security of symmetric keys that are shared among payment participants to protect personal identification numbers (PINs) and other sensitive data. Key Blocks are immune to specific types of attacks and also restricts further key usage to enhance security.

Q. Which PCI Security Standard requires the implementation of Key Blocks?

The PCI PIN Security Requirements and Testing Procedures (PCI PIN Standard) require the implementation of Key Blocks:

- Requirement 18-3: *Encrypted symmetric keys must be managed in structures called key blocks. The key usage must be cryptographically bound to the key using accepted methods.*

Q. When were Key Block requirements first introduced?

Key Blocks were first introduced in version 2.0 of the PCI PIN Standard, published in December 2014, to increase security for encrypted keys.

Q. Does the PCI PIN Standard apply to both conveyance and storage of keys?

Yes. The PCI PIN Standard applies to both conveyance and storage of keys.

Q. Does the PCI PIN Standard apply to only TDEA/TDES keys?

No. The PCI PIN Standard does not only apply to TDEA/TDES keys. Both AES and TDES keys are required to be managed in key blocks as stipulated by ANSI X9.

Q. Who must comply with the PCI PIN Standard?

Customers and their agents (e.g., key-injection facilities and certificate processors) performing PIN encipherment or any other aspect of PIN processing involving PIN entry must comply with the PCI PIN Standard and all other applicable PCI PIN Security Standards.

Q: Is PCI PIN validation to Mastercard required?

No. PCI PIN validation to Mastercard is not required.

Q. Will Mastercard grant an extension for entities not compliant by the dates outlined in PCI PIN security requirement 18-3?

No. As PCI PIN validation to Mastercard is not required, Mastercard will not grant an extension for entities not compliant by the dates outlined in PCI PIN security requirement 18-3.

Q. When must customers and their agents be compliant with PCI PIN security requirement 18-3?

Customers and their agents are required to comply with the following phased Key Block implementation dates:

- **Phase 1** Implement Key Blocks for internal connections and key storage within service provider environments. This includes all applications and databases connected to hardware security modules (HSMs). Effective date: 1 June 2019. **Completed**
- **Phase 2** Implement Key Blocks for external connections to associations and networks. New effective date: 1 January 2023 (replaces previous effective date of 1 June 2021).
- **Phase 3** Implement Key Block to extend to all merchant hosts, point-of-sale (POS) devices and ATMs. New effective date: 1 January 2025 (replaces previous effective date of 1 June 2023).

Q. Is the date for phase 1 (implement key blocks for internal connections and key storage within service provider environments) already effective?

Yes. The date for phase 1 was effective 1 June 2019. This date has remained unchanged.

Q. Can the Mastercard network currently support the use of the new key block format in the production environment? Is Mastercard ready?

Yes. The Mastercard network can support the use of the new key block format in the production environment. Mastercard has been ready since January 2020.

Q. Are current ANSI X9.17 key exchanges compliant with PCI PIN security requirement 18-3?

No. Current ANSI X9.17 key exchanges are not compliant with PCI PIN security requirement 18-3. ANSI X9.17 key exchanges must be converted to key blocks under phase 2 of Key Block implementations by 1 January 2023.

However, Mastercard will still allow customers to continue using ANSI X9.17 key exchanges after the effective date, 1 January 2023.

Q. Where can customers and their agents find more information on key exchanges and testing requirements to support PCI PIN security requirement 18-3 involving Key Blocks?

Customers and their agents can find more information on key exchanges and testing requirements to support PCI PIN security requirement 18-3 involving Key Blocks by referring to Mastercard's *2117—Introducing Cryptographic Key Block Changes to Support Phase 2 of PCI Mandates* available on [Mastercard Connect™](#).

Q. Who should customers and their agents contact with questions on operational requirements for key management use?

Customers and their agents should contact Customer_Support@mastercard.com with questions on operational requirements for key management use.

Q. Can “equivalent methods” for Key Block compliance be used?

Yes. In July of 2019, the PCI SSC added FAQ Q39 to the [PCI PIN Technical FAQs version 3](#) detailing how an independent expert may validate the integrity of a processed solution and the qualifications that the independent expert must possess (interoperable methods include those defined in ANSI TR-31 and International Organization for Standardization [ISO] 20038).

Q. Does Mastercard allow for “equivalent methods” for Key Block compliance to be used?

Yes. Mastercard is supporting Accredited Standards Committee (ASC) X9 TR-31 version ID “B” for online key exchanges. Customers and their agents may store the keys in any compliant manner. For more information on Mastercard’s requirements for Key Blocks, customers and their agents should contact Customer_Support@mastercard.com.

Q. What resources does PCI SSC offer to support the implementation of Key Blocks?

- [Key Blocks Blog Series](#)
- Information Supplements: [Cryptographic Key Blocks: PIN Security Requirement 18-3 – Key Blocks](#)
- [PCI PTS PIN Security Requirements Technical FAQs version 3](#)